

Ruth George

CONTACT

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RESEARCH PROJECTS

Use Cases of Flare Clustering in Machine Learning

2026

- Comparing flare clustering to spectral clustering to explore possible benefits to this new method
- Working on various dimensionality reduction and clustering techniques to further explore use cases of flare clustering

Probability of Language Comprehension Based on Levenshtein Distance

2025

- Created a reaction-advection-diffusion model to predict language comprehensibility over space and time of Hejazi Arabic in Saudi Arabia
- Conducted parametric study to determine the most important factors
- Compiled reliable data from various sources and programmed a solution for the equations using finite element method

Predicting Federal Funds Interest Rate using Markov Chains

2024

- Created a Markov chain using current interest, inflation, unemployment, and GDP growth rates to predict federal funds rates quarterly
- Benchmarked results against Q1 and Q2 2024 and predicted Q3 and Q4 2024

Instructional AI Assistant

2023

- Created an AI product that helps teachers give feedback on student performance using Python and Panda libraries
 - Interviewed teachers and used simulated data of student performance
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EXPERIENCE

Intern | Snapology Los Gatos

2024

- Taught students to build robots using block coding and legos
 - Provided leadership, guidance, motivation and instruction to the students during each activity period
 - Enforced safety precautions and increased student interest in robotics
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EDUCATION

UC San Diego | 2025–present

- Regents Scholar
- GPA: 4.00

Archbishop Mitty High School | 2021–2025

- GPA: 4.76
- Principal's Honor Roll 21–25
- Math Department Award 22, 23, 25